

Multi – level Linked List – Class Tweaks

This is standard declaration of Constructor

```
Node(int val)
{
    data = val;
    next = nullptr;
    child = nullptr;
}
```

When we write assignments inside the {} brackets, C++ actually does the work twice:

1. First, it creates the variables (data, next, etc.) and fills them with default "garbage" memory values.
2. Then, it goes into the constructor body and overwrites those garbage values with your requested values.

But this can be avoided using Member Initializer List or just initializer List.

It is the standard, most efficient way to give initial values to your class variables (members) when an object is created.

Here is exactly how it breaks down:

```
Node(int val) : data(val), next(nullptr), prev(nullptr),
child(nullptr) {}
```

- **Node(int val)** – This is the constructor signature. It says, "When creating a Node, expect an integer called val."
- **:** – The colon is the magic symbol. It tells the compiler, "The initialization list is starting now."
- **data(val)** – This directly initializes the class variable data with the value of val.

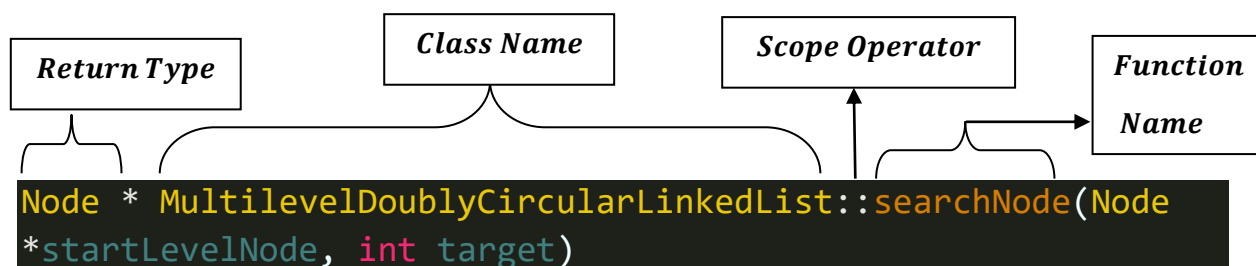
- `next(nullptr)` – This directly initializes the `next` pointer to `nullptr`. (Same for `prev` and `child`).
- `{ }` – This is the actual body of the constructor. Because we already set all the values in the list, the body is completely empty!

And we see:

```
class MultilevelDoublyCircularLinkedList
{
private:
    Node *head; // Encapsulated Head Pointer

    // --- Helper Declarations ---
    Node *searchNode(Node *startLevelNode, int target);

Node *MultilevelDoublyCircularLinkedList::searchNode(Node
*startLevelNode, int target)
{...}
```



Here Node is a Class Name , represents pointer ,hence it i. e. hold memory address of class Node whose return type is a pointer variable ,here : startLevelNode which holds memory address just like struct.*

Now what does this mean?

```
MultilevelDoublyCircularLinkedList::searchNode
```

searchNode is a function declared under class name

MultilevelDoublyCircularLinkedList , now its getting defined , and searchNode is under class

`` MultilevelDoublyCircularLinkedList`` is determined by scope operator ::

- **Node *** → The return type (it returns a pointer to a Node).
 - **MultilevelDoublyCircularLinkedList** → The name of the Class.
 - **::** → The Scope Resolution Operator (translates to "owns" or "belongs to").
 - **searchNode(...)** → The actual function being defined.
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